Manideep Emmadi

Hyderabad, India

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Education

B.Tech in Electronics and Communication Engineering **2024 – 2027**
Methodist college of Engineering and Technology *India*

Diploma in Electronics and Communication Engineering **2021 – 2024**
Mahaveer Institute of Science and Technology *India*

Relevant Coursework

- Digital Electronics
- Signals and Systems
- Microprocessors
- Semiconductor Physics
- Analog Electronics
- VLSI Design
- Embedded Systems
- Control Systems

Experience

DRDO – Research Centre Imarat (RCI) **2025**
Electronics / VLSI Intern *Hyderabad, India*

- Applied digital design concepts and hardware verification methodologies in simulation environments.
- Developed and executed Verilog testbench structures for functional validation.
- Analyzed semiconductor design workflows used in defense electronics systems.

Techieyaan **2023**
Internship Trainee *Hyderabad*

- Completed structured training in electronics and embedded systems fundamentals.
- Executed technical assignments involving IoT concepts and hardware interfacing.
- Implemented Arduino-based systems and basic circuit design applications.
- Collaborated with peers on technical modules and mini engineering projects.

Projects

16-bit ALU Processor Design | *Digital Design, Circuit Simulation*

- Designed and implemented a 16-bit Arithmetic Logic Unit supporting multiple arithmetic and logical operations.
- Conducted power and transient analysis to evaluate circuit efficiency.
- Evaluated timing performance and operational behavior of processor components.

Traffic Light Controller FSM | *Verilog, Xilinx Vivado*

- Developed a Verilog-based finite state machine (FSM) for traffic signal control.
- Simulated and verified system functionality using Xilinx Vivado.
- Optimized state transitions to improve signal sequencing efficiency.

Multi-Baud Rate UART Transmitter | *Verilog HDL, Xilinx Vivado*

- Designed a synthesizable UART transmitter utilizing a 6-state Moore Finite State Machine (FSM).
- Implemented configurable baud rate generation supporting dynamic frequency scaling (DFS) for low-power operation.
- Validated precise serialization of data frames and state delays using self-checking testbenches in Vivado.

Electrostatics Simulation | *Python, NumPy, Matplotlib*

- Developed a Python simulation to visualize electric field distribution using numerical methods.
- Implemented scientific computation and visualization using NumPy and Matplotlib.

Technical Skills

Programming Languages: Verilog, Python, C
Hardware / Embedded Platforms: Arduino, Sensor Interfacing
EDA & Development Tools: Xilinx Vivado, Arduino IDE, VS Code, Custom Compiler
Simulation & Analysis: Circuit Simulation, Power Analysis, Transient Analysis
Libraries: NumPy, Matplotlib
Domains: VLSI Design, Digital Electronics, Embedded Systems, IoT Systems

VLSI & Hardware Skills

HDL: Verilog
Digital Design Concepts: Finite State Machines (FSM), ALU Design, Digital Logic
EDA Tools: Xilinx Vivado
Circuit Analysis: Power Analysis, Transient Analysis
Hardware Platforms: Arduino, Sensor-based Systems

Leadership

Street Cause **2024 – Present**
Divisional President

- Directed divisional initiatives and coordinated volunteers for community service programs.
- Secured over 10,000 in fundraising as part of the RFC 11 campaign.
- Organized and managed social impact initiatives and outreach events.